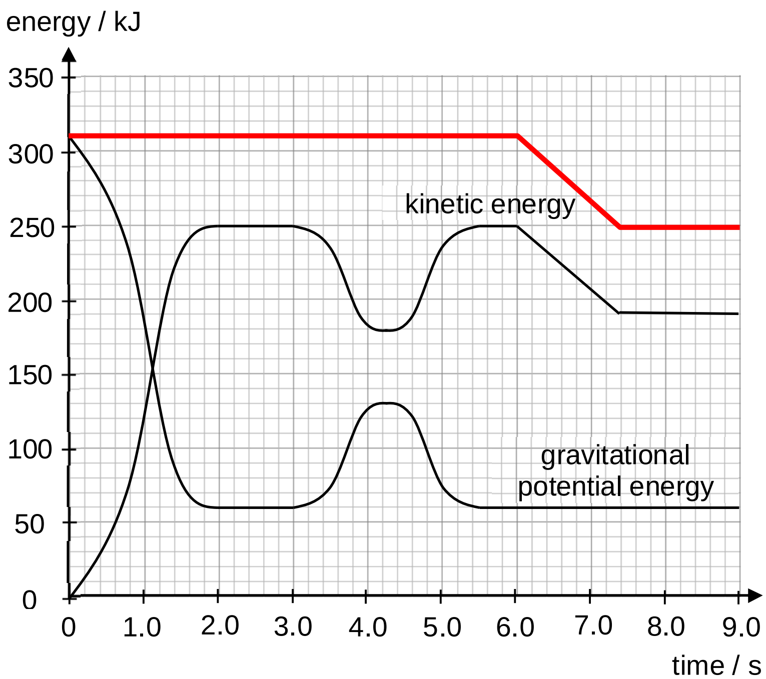
No.	Solution	Remarks
2(a)		[1] horizontal line from $t = 0$ s to $t = 6.0$ s, with mechanical energy at 310 kJ [1] straight line joining points (6.0, 310) and (7.4, 250), followed by horizontal line from $t =$ 7.4 s to $t =$ 9.0 s, with mechanical energy at 250 KJ
2(b)	From the Fig 2.2, $t = 3.0$ s to $t = 4.3$ s, the increase in gravitational potential energy is $\Delta GPE = 130 - 60$ $= 70 \text{ kJ}$ $mg\Delta h = 70000$ $\Delta h = \dfrac{70000}{380 \times 9.81}$ $= 18.778 \approx 19 \text{ m}$ The increase in height, and therefore the diameter d of loop, is 19 m.	[1] for correct substitution [1] for correct answer
2(c)(i)	start time = 6.0 s end time = 7.4 s	[1] for correct start and end times.
2(c)(ii)	rate of loss of KE $= \dfrac{\text{loss of KE}}{\text{duration in zone A}}$ $= \dfrac{(250 - 190) \times 10^3}{7.4 - 6.0}$ $= 42857 \text{ W} \approx 43000 \text{ W}$	[1] for correct substitution [1] for correct answer
2(c)(iii)	From Fig 2.2, the car enters zone A with a KE of 250 kJ,	

No.	Solution	Remarks
	$\frac{1}{2}mv^2 = 250\ 000$ $v = \sqrt{\frac{2(250\ 000)}{380}}$ $= 36.274 \approx 36.3$ m s^{-1}	[1] for correct substitution [1] for correct answer
2(c)(iv)	Using $P = Fv$, $F = \frac{P}{v}$ $= \frac{42\ 857}{36.274}$ $= 1181.49$ ≈ 1200 N	[1] for correct substitution [1] for correct answer
3(a)	Internal energy is determined by the state of the system and it can	[1] for correct